15. An ideal gas, contained in a cylinder by a frictionless piston, is allowed to expand from volume V_1, at a pressure p_1, to volume V_2, at pressure p_2. Its temperature is kept constant throughout. The work done by the gas is

A Zero, because it obeys Boyle's Law and therefore $p_2V_2 - p_1V_1 = 0$.

B Negative, because the pressure has decreased and so the force on the piston has been diminishing.

C Zero, because it has been kept at constant temperature and so its internal energy is unchanged.

D Positive, because the volume has increased.